

established at each of these locations. These RF fingerprints are then stored in a database and used later at runtime to match the currently observed RF environment to the previously stored data. Using pattern-matching and a series of advanced locationing algorithms, the system can then determine the current location—with reliable “room level” accuracy (approximately 3 meters)—of a device or person with either the 802.11 NIC or an 802.11 asset tag. Location information is used by 802.11 active RFID asset tracking systems to display the current location of different types of equipment as well as enable users to search for specific assets.

These newer approaches offer advantages over traditional active RFID technologies. First, they use the existing 802.11 WLAN infrastructure which offers significant cost reduction in terms of initial installation and on-going operation/maintenance. Wireless asset-tracking capabilities have traditionally been the domain of specialized Real Time Tracking Systems (RTLS) that require their own dedicated, proprietary network infrastructure. These single-purpose tracking networks—due to their hardware-intensive deployment—have proven to be costly to purchase, install, implement and maintain. Wi-Fi-based Enterprise Asset Visibility solutions leverage an organization’s existing Wi-Fi infrastructure, and eliminate the need for specialized radio receivers, access points, antennas and wiring. By deploying on the existing Wi-Fi network, using software-only location architectures, the newer generation of Enterprise Asset Visibility solutions dramatically lowers the total cost of ownership and opens up new possibilities by combining services together. Second, they enable opportunities to deliver exciting new applications that combine voice, data and location awareness which can improve efficiency and customer service. In sum, these new approaches represent a superior, more efficient and cost-effective alternative to first-generation RFID technology that require enterprises to deploy separate, single-purpose networks of proprietary readers, cabling and tags in order to provide location/tracking capabilities.